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Predicting Students at Risk of Early Dropping Out from Course Using Ensemble Classification Methods

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Abstract

Many research studies confirmed that the application of machine learning methods for predicting students at risk of academic failure can be beneficial for improving student performance in higher education institutions. Simultaneously, these studies emphasised the importance of the input data quality, its precise pre-processing and thoroughly selected features on final performance metrics of the applied classification algorithms. However, the stand-alone application of well-known classification algorithms still has performance limitations, which can be outperformed using ensembles of classifiers. Therefore, this paper aims to evaluate the overall contribution of the selected ensemble methods to the early prediction of students at risk of dropping out of learning courses. All phases of CRISP-DM methodology are described in this study to explain the complexity of the problem of the practical adoption of voting-based heterogeneous ensemble methods in the case of data obtained from the course of the university learning management system compared to traditional classification algorithms enhanced by ensemble learning with AdaBoost and XGBoost methods. As a result, the study confirmed that the features which characterise the students' performance and their interactions in the course can be considered the most significant for the identification of the students at risk of dropping out. Moreover, in the case of advanced ensemble algorithms, the overall accuracy, recall, precision, and f1 score were improved by 2-4%. The proposed method has demonstrated exceptional performance. Finally, the paper summarises the findings and their implications for effective prediction.

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1. Introduction

Blended learning provides students with a flexible educational model and a customised approach by combining traditional classroom instruction with online learning activities. It is frequently utilised in higher education because it enhances students' course involvement and increases students' chances of achieving academic success. However, because blended learning is so complicated, it can be challenging for teachers to keep an eye on their students' behaviour and offer timely assistance when necessary [1] [2]. For almost ten years, researchers have been attempting to build a solution through the use of data mining (DM) and machine learning (ML) techniques that can analyse and predict students' performance as well as the underlying cause of it [3].

According to Freitas et al. [4], learning analytics has the potential to be an effective method for fostering the academic achievement of students. It is the measurement, collection, analysis, and reporting of data about learners and their contexts to understand and optimise learning and the environments in which it occurs. It is possible to use it to track individual students' engagement, attainment, and progression in a manner that is very close to real time. In addition to this, it may be used to conduct an analysis of the features of the learning behaviours of students, as well as to anticipate their academic progress and facilitate prompt intervention. Traditional assessment tools such as questionnaires, examinations, and classroom-based observations, according to Klein and Hess [2], can be improved with the help of learning analytics through formative and direct measures derived from visible, timely, predictive, and newly collected data.

Therefore, it is crucial to have accurate predictions of students at risk of academic failure to implement timely instructional interventions to improve student performance and the efficiency and effectiveness of higher education [5]. In the past ten years, there has been a considerable uptick in the number of studies that attempt to predict student performance. These predictions place the majority of their emphasis in the classification and regression models. Predicting a final grade or score is less common than using the classification (pass/fail). Various classification techniques, such as random forest, fuzzy logic, k-means clustering, naive Bayes, decision tree, support vector machines, artificial neural networks, and k-nearest neighbour, have each produced remarkable results on a variety of topics. The fact that each student's data set is unique explains why there are bound to be some differences in their accomplishments [6], [7], [8], [9].

The same algorithms can have varying results when applied to various data sets. Furthermore, each algorithm has various biases based on the type of data it is applied to, which might make it difficult to establish which algorithm is the one that is universally accepted. Because of this, it is recommended to employ ensemble learning models, which incorporate the predictions of various algorithms, to create more accurate predictions, to surpass the generalisation capability of a single learning algorithm, and to exceed the robustness of a single learning algorithm. In addition, studies need to be carried out using up-to-date machine learning algorithms, and the analysis process needs to consider learning logs derived from large sample groups [3].

This research aims to perform an ensemble model that might identify students likely to struggle academically during the course. Within the scope of this investigation, attempts were made to find responses to the following questions:

- Which ensemble model is the most accurate in identifying students at risk of failing their classes?
- Which characteristics of the students impact the most on the accuracy of the prediction?

This paper's proposed research contributes to the existing literature on predicting students' academic performance. In particular, it evaluates different ensemble models to improve the accuracy of student performance prediction using data features available within a 13-week academic course. The structure of the research paper is as follows. Initially, a summary of prior research on predicting student success using machine learning techniques in the domain of learning analytics is presented. Then, beginning with Section 3, the individual phases of the CRISP-DM methodology, which manages the knowledge discovery process, are progressively introduced. This section describes the primary characteristics of the investigated dataset. Different ensembled machine learning classifiers have also been applied to

the student attrition prediction models presented in this section. Section 4 addresses the derived results and evaluation of the models based on various performance measures. Finally, section 5 concludes with a summary of the research findings.

2. Related Works

Several studies have approached predicting students' academic achievement as either a regression problem [10] or a classification challenge [11]. The results of the learners' perspectives were predicted in the first category, while the learners' outcomes were estimated as either a pass, a failure, or a dropout in the second category. In addition to the utilisation of a variety of data analysis strategies, identifying factors that affect student success was also carried out to forecast student performance. In addition, the study areas of predicting student performance can be evaluated from a number of different points of view. These include predicting early withdrawal from ongoing courses, analysing inherent aspects influencing their achievement, and applying statistical methods to evaluate student achievement. The idea of making predictions as early as possible has only recently emerged in this field. It comprises methods for assessing students in real time to retain them by offering suitable procedures and interventions, then monitoring and decreasing the failure rates that occur as a result.

Research [11] analysed the learning habits of students and attempted to determine whether students were at risk of failing. These analyses made use of machine learning approaches. The research used a subsequent strategy to turn the class time into a series of weekly structures and to measure the learning achievement of students based on their interaction with virtual learning environments (VLEs). Attendance, quizzes, and assignments were utilised as predictors of students who may be at risk, with the inclusion of midterm examinations in the ninth week [12]. Machine-learning algorithms were employed to make these predictions. In a different piece of research, multiple data mining methods, including the decision tree, the naive Bayes classifier, the k-nearest neighbour, the support vector machine (SVM), and the multi-layer perceptron (MLP), were used to predict students who were at risk. Multiple studies used logistic regression as their base model and then compared each other to determine which model provided the most accurate predictions [13].

2.1. Ensemble learning

The usage of ensemble learning is advised for increasing the stability of a single learning algorithm and the accuracy of its prediction. According to [14], the ensemble learning model is predicated on a meta-algorithm that creates a final prediction by combining numerous models of the same or different types that have been individually trained. The many kinds of fundamental learning algorithms incorporated into a model and the processes used to develop the model are two of the criteria that may be used to categorise ensemble learning. Ensemble learning can be heterogeneous or homogeneous depending on the many kinds of learning algorithms incorporated into the model. In the heterogeneous ensemble model, the same training data is fed into a variety of learning algorithms, or it is fed into the same learning algorithm but with a variety of parameter values. According to [15], the homogeneous method involves segmenting the initial training data into a variety of sub-datasets before applying that information to the same learning algorithm in accordance with the number of sub-datasets. Second, ensemble learning can be broken down into two distinct subcategories: boosting methods and averaging approaches. Learning models are developed individually with the use of averaging procedures. In order to minimise the amount of variation, the predictions that these simple models generate are averaged. The primary training data is typically segmented into several different random subsets when using boosting approaches. These subsets are used to train the samples of the same basic learning algorithm selected earlier. The results of the collected forecasts are then merged to produce a conclusive prediction. Any type of learning algorithm is compatible with the application of boosting strategies. On the other hand, they typically function most effectively when combined with sophisticated and robust algorithms.

In recent years, academics have been using many different ensemble models to predict the achievement of students. The goal of the study [16] was to bridge the gap between the empirical prediction of student performance and the available machine-learning approaches in the context of a remote education setting. They suggested a voting-based online ensemble of classifiers incorporating an incremental version of DT, NB and the ANN algorithm. All three of these algorithms work together to produce a classification. Research [3] provided a methodical technique for selecting

an appropriate ensemble learner from various machine learning algorithms. This method was based on the Gini index and the p-value. They studied two independent datasets, one at the 20% stage and the other at the 50% level of the course delivery. The results of the experiments indicated that the ensemble models were able to reach a high level of accuracy while maintaining a low rate of false positives (FP) throughout all stages for both datasets.

3. Methodology

CRISP-DM (Cross-Industry Standard Process for Data Mining) is among the most popular knowledge discovery processes. As the creation of predictive models in this paper is a typical knowledge discovery problem endeavour, the CRISP-DM methodology was also utilised. This study introduces a student performance model by utilising voting-based ensemble approaches. Ensemble methods are a form of approach to problem-solving that involve the utilisation of multiple models in order to find a solution. In contrast to classical learning approaches, which train data using a single learning model, ensemble methods aim to train data using a variety of models and then combine them to take a vote on their outcomes. This contrasts traditional learning approaches, which only use a single learning model to train data. The accuracy of predictions made by multiple models working together, often known as an ensemble, is typically higher. This strategy seeks to achieve the objective of providing an accurate assessment of the qualities that, to a certain extent, influence the academic success of a student. Fig. 1 is an illustration of the primary steps that comprise the suggested methodology.

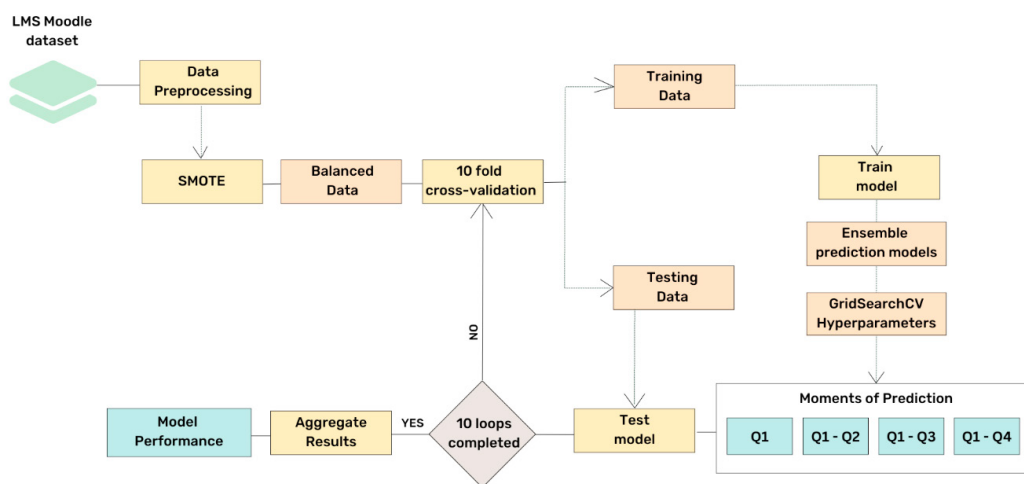


Fig. 1. Proposed methodology.

Python 3.8 was used throughout all stages of the proposed methodology, and the platform used was the open-source and free Anaconda distribution. Scikit-learn was used for machine learning methods. Pandas were used to import, construct, and manipulate Data Frames. NumPy was used for array computing. Matplotlib and Seaborn were used for data visualisation. Imblearn was used due to an imbalanced dataset. We used the SHAP (SHapley Additive exPlanations) to visualise and better interpret the findings. 10-fold cross-validation is used to address the issue of overfitting.

3.1. Business understanding

This study explored the associations between student accomplishment and online activities of 110 university students who took a blended course, which combined traditional lectures and lab sessions with online interactions and online access to course materials. All students participating in this study were in Computer Science study program during the winter semesters from September 2020 to January 2022. The course was available through the LMS Moodle. In this environment, students can access online content, engage with one another on projects, communicate

with the instructors, hand in homework, participate in quizzes, and assume overall responsibility for their education. The activities of the students are documented in the log file of the system. It provides information about the identity and IP address of each student, as well as the time of the visit and the activity that was carried out.

In one of our studies [17], we attempted to predict students' dropout at the course level using traditional classification algorithms. The accuracy of predictions ranged between 77% and 93%. Limitations of the study include the small size of the dataset and the number of independent features. In addition, selecting the time threshold for which the prediction was made represents an additional vulnerability. Due to the absence of a direct time variable, it was necessary to identify student performance milestones that were challenging to predict.

Following on from the above, we are trying to investigate other methods and take into account the time in which the prediction would be meaningful.

3.2. Data understanding

The data used in this study has changed due to the pandemic situation, which has brought about significant adjustments to how content is presented and the mode of instruction through the LMS Moodle. The course contained numerous learning-based resources and interactive activities encouraging students to become more active and acquire knowledge and skills.

Each course at the university is divided into 13 weeks by default. Using this information, various moments of prediction can be studied throughout the course. It is possible to construct a framework containing a prediction model for each day following the course's start. However, for the sake of simplicity, it is determined to divide the course timeline into four evenly distributed moments of prediction. The first quarter (Q1) comprises all student interactions through the end of the period. The second quarter (Q1-Q2) data includes the newly recorded interactions through the course's midpoint in addition to the first quarter data. The same logic applies to the third (Q1-Q3) and fourth quarters (Q1-Q4). As no one desires prognostic moments on the final day of the course, the end of the fourth quarter is moved one week prior. In other words, the recorded student interactions from the first quarter, second quarter, and third quarter are accessible until seven days prior to the last day of the course during the fourth quarter.

Table 1. Number of feature distributions for each examined period with the student's final outcome.

Time	Number of features	Distribution
Q1 (1 – 3 week)	7 (AT, Z1-Z5, A1)	
Q2 (4 – 6 week)	12 (AT, Z1-Z9, A1, A2)	Pass: 83 (75,46 %)
Q3 (7 – 9 week)	17 (AT, Z1-Z13, A1, A2, T)	Fail: 27 (24,54 %)
Q4 (10 – 12 week)	21 (AT, Z1-Z17, A1-A3, T)	

Table 1 shows the distribution of the dataset into individual quarters together with the number of features and the overall distribution of the student's final outcome. The imbalance between the number of passing (83) and failing (27) students biases the predictive model. We needed to apply a careful data balancing technique to ensure better precision without compromising the learning value of the data. We chose the Synthetic Minority Oversampling Technique (SMOTE) [18] to create extra data points in the training data set to balance the data classes.

After the student's participation in the courses had ended, the reporting function of Moodle was utilised to export the data from the system's log file into .csv files. The first resources were made available to students at the beginning of the courses, and those materials continued to be updated weekly until the end of the semester. The data set included two distinct kinds of information: student achievements during the course and interaction data on students accessing materials and participating in activities within the learning management system. An obvious question is how the points obtained from these activities and the student's behaviour throughout the term can effectively predict the success or failure of students.

3.3. Data pre-processing

The first step in pre-processing is data cleaning, which consists of techniques for resolving issues with missing data, removing inaccurate and inconsistent data, and identifying outliers in order to improve data quality.

As part of our study, we utilise two files. The first file contains academic records of activities in the course, as well as information on how many attempts the student had in the course being taught - the variable *AT* (Attempt Count). The second file contains the remaining entries representing various course-related activities that students could fulfil, such as *Z* (Assignments), *A* (Tests) and *T* (Exam). The accompanying Fig. 2 shows the complete dataset with merged files.

This dataset had no records with inconsistent, missing, or outlier values. Consequently, no records were removed during the data cleaning procedure. The next step in the pre-processing is data scaling. Research has shown that machine learning algorithms are trained effectively when data sets are normalised or scaled, resulting in better prediction and faster processing or training. Due to the ensemble machine learning model's reliance on tree algorithms unaffected by the range of values, scaling only had to be considered for the SVM and KNN algorithms. A min-max normalisation was selected for both algorithms to adjust the values to the range [0, 1] based on recommendations. Table 2 describes all the characteristics of the learning activities.

Table 2. Description of each learning feature.

Feature Code	Description
AT (Attempt Count)	The number of attempts made by each student in a given course.
Z1 – Z17 (Assignments 1-17)	Successful completion of every available assignment.
A1 – A3 (Tests 1-3)	The score obtained throughout all three tests taken during the semester.
T (Exam)	The results of the midterm exam.

3.4. Ensemble prediction model

This research introduces a student performance model by utilising ensemble approaches. In contrast to classical learning approaches, which train data using a single learning model, ensemble methods aim to train data using a variety of models and then combine them to take a vote on their outcomes. This contrasts traditional learning approaches, which only use a single learning model to train data. As a result, the accuracy of predictions made by multiple models working together, often known as an ensemble, is typically higher.

Basic and advanced boosting ensemble methods are used in this paper to objectively evaluate the influence attributes on the achievement standards of students and to enhance the execution of the student's prediction model. To build our basic prediction models, we relied on five widely-used methods: logistic regression (LR), random forest (RF), decision tree (DT), k-nearest neighbours (KNN) and support vector machine (SVM). In addition, AdaBoost and extreme gradient boosting (XGB) methods were selected for the advanced prediction models. All models were running with optimal hyperparameters settings obtained through the GridSearchCV technique.

The term boosting is used to describe a family of methods that can improve the performance of weak learners. The popular boosting method is straightforward: after training a series of classifiers in sequence and receiving their predictions, the method focuses on minimising the errors of the weakest learner by adjusting its weights. The AdaBoost method is able to overcome this adaptive constraint. The core idea behind the algorithm is to focus primarily on challenging categorisation patterns. The interest rate is the equal weight given to each subset in the training data. Each iteration magnifies the impact of incorrect classifications while diminishing the weight of correct ones. AdaBoost then pools the learners to improve the weakest classifiers and produce a powerful learner [18].

In boosting, the parameters of each classifier are calculated with the help of the output of the previous classifier. Then, each case is given a probability calculated based on how much emphasis it should be given. All of the classifiers mentioned above were trained with 10-fold cross-validation. This method splits large datasets into ten equal-sized subsets, nine used for training and one for testing. The average error rate of the test models is then computed after the procedure is repeated ten times.

3.5. Evaluation measures

The created and trained models from the previous phase will be evaluated using the defined metrics in this phase of the CRISP-DM methodology. The quality of the output from any classification algorithm is greatly affected by the

type of data used as input. After creating and training the model, its performance can also be evaluated on the training data, which can be misleading. However, the goal should always be to create a model that can best predict new observations on which it is not trained. In the case of models that evaluate the training data too well, the success of the classifier most likely does not correspond to reality, as it was specialised and trained on this data. Such a phenomenon is referred to as model overfitting. The final model must be robust enough to correctly classify test data outside the training set without significant performance differences. Therefore, it is always desirable to evaluate the success of the test data set and how much the classifier encounters a so far unknown data set. There are several metrics according to which the success of the given classifier can be determined. Overall accuracy, recall, precision and fl score are the most frequently used metrics for evaluating classification models.

Table 3. Classification performance metrics.

Metric	Explanation	Formula
Accuracy	The percentage of correctly predicted results.	$(TP + TN)/(TP + TN + FN + FP)$
Precision	The percentage of correct positive observations.	$TP/(TP + FP)$
Recall	The percentage of positives that are correctly predicted as positive.	$TP/(TP + FN)$
F1-score	The harmonic mean of precision and recall.	$2 * (Precision * Recall)/(Precision + Recall)$

4. Results

The proposed methodology was evaluated using the Moodle LMS dataset with 110 rows and 22 columns. The objective of these four distinct binary classification problems was the early prediction of students at risk of failing the course, students with high academic achievements, and dropouts. Based on this information, four prediction moments were constructed, representing the first quarter of the course, the midterm, the third quarter of the course, and one week prior to the course's conclusion. Two prediction models have been developed to test the proposed framework: basic and advanced boosting ensemble methods.

Multiple variables may have a direct or indirect relationship with the learner performance model. Each model has been evaluated with respect to multiple prediction moments, four quarters as the binary classification problems.

Applying basic ensemble methods with traditional classifiers (LR, RF, DT, SVM, KNN) yielded favourable results compared to the advanced level, where AdaBoost and XGB were utilised. Each ensemble trains with each classifier and then accumulates the results using a majority vote rule to improve the student's model's prediction performance.

Table 4. Summary of ensemble model's performance using a 12-week data set.

Classification Model	Basic ensemble methods LR, RF, DT, KNN, SVM				Advanced ensemble methods AdaBoost, XGB			
	Q1	Q1-Q2	Q1-Q3	Q1-Q4	Q1	Q1-Q2	Q1-Q3	Q1-Q4
Accuracy	85.30%	88.24%	91.18%	94.12%	88.49%	91.34%	94.24%	97.06%
Recall	0.87	0.89	0.92	0.92	0.90	0.91	0.93	0.95
Precision	0.86	0.88	0.90	0.94	0.89	0.92	0.95	0.97
F1 score	0.85	0.87	0.91	0.93	0.88	0.90	0.94	0.96

To ensure that the ensemble models were reliable, we subjected them to a 10-fold cross-validation procedure. The model is trained with 90% of the data and evaluated with 10% across 10 iterations. It's worth mentioning that each time the 10% is split for testing, a different set of data points is chosen.

Table 4, with the macro-averaging of precision and recall, show that both ensemble methods work well for both classes. As time progressed, the prediction models became more accurate as they started with 85.30% at the end of the first quarter and ended with 94.12% seven days before the end of the course. The AdaBoost with XGB achieved better accuracy 88.49%, 91.31%, 94.24% and 97.06% for individual quarters. Although accuracy is the most employed performance metric for evaluating model results, relying only on accuracy could lead to a misunderstanding because

the model can only accurately predict one class. In order to address this issue, this study also considers precision, recall, and F1-score as evaluation metrics. These indicators play an important role in class prediction and provide greater insight into prediction outcomes. The overall precision, recall, and F1-score results for advanced methods are 0.93, 0.92, and 0.92, which outperforms the base models (0.90, 0.90 and 0.89). The recall results of 0.93 for the advanced ensemble methods on the testing set indicate that the model can predict nearly 92% of dropouts.

Similarly, a precision value of 0.93 indicates that we are approximately 93% accurate in predicting successful students after course delivery. The fact that basic ensemble methods can identify both classes with an accuracy of 90% demonstrates that each class can also be efficiently managed. Regarding the F1 score, its values range from 0 to 1, where 0 indicates extremely low results and values close to 1 indicate effective results. Our technique achieved an aggregate F1-score of 0.92 and 0.89. The advanced method's superior prediction performance indicates that it is a viable option for predicting university student dropout.



Fig. 2. Accuracy of a specific model within a) basic ensemble method (left), b) advanced ensemble method (right).

In addition, the performance of individual algorithms in both approaches was examined. According to Fig. 2, the random forest classifier outperforms the other classifiers. The difference in accuracy between each model is either more or less than 1-3%. An exception was the KNN algorithm, which is almost 10% worse compared to RF at a certain moment of prediction. Each algorithm had several essential parameters to optimise, and a systematic grid search was implemented to improve the accuracy of results. However, RF has significantly more hyperparameters than other methods, contributing to its superior performance. In contrast, Adaboost and XGB are relatively comparable in performance, so we can consider them suitable for the proposed problem.

Feature importance is prior to developing the ensemble model to determine student performance. During this phase, the optimal data set features were identified and selected for training all candidate learning algorithms to generate the final ensemble models. Due to the large number of variables (22), not all may be useful for classification. Therefore, the technique for determining the significance of individual variables was employed in this study. We used the SHAP method for visualisation and better interpretation of the findings. It explains how each input variable affects the model and allows analysis of the data set and the given problem but does not evaluate the quality of the prediction itself. We visualised the importance of variables using the SHAP method, as shown in the following Fig. 3. The graph shows the ordering of the input variables in order of their influence on the prediction for the AdaBoost classifier in each quarter. We can see from the bar charts that all input variables contribute positively to the final prediction. In the first period of Q1, the test (A1), together with the last two assignments (Z4, Z5), have the most significant influence, followed by the number of attempts (AT) of the student in the given course. Within the second examined period (Q1-Q2), both tests (A2, A1) proved to be the most important, followed by the most current assignment (Z9) in this period. The penultimate examined section (Q1 - Q3) again indicated the most critical features, which included the mid-term exam (T), together with tests (A2, A1) followed by some assignments (Z5, Z8). For the entire period (Q1 - Q4), visualisations proved to have the most decisive influence of all tests (T, A1-A3), followed by the most current assignments (Z11 - Z16) that students could submit in this period and their attempt count (AT) in the course. Similar findings were also found with other algorithms.



Fig. 3. Feature importance with SHAP for AdaBoost classifier within each quarter.

5. Conclusions

Vast quantities of data are generated in learning systems. This data contains concealed information that could be used to improve students' academic success ensemble methods were used exclusively. This study used voting-based ensemble methods exclusively to construct a model for predicting student achievement. The basic predictive model comprises classifiers such as LR, RF, DT, KNN and SVM, and the boosting method is used to improve the performance of these classifiers. The retrieved results demonstrate that these models are almost as good as advanced classifiers. The proposed advanced ensemble method then combines two classifiers: AdaBoost and XGB. These methods produced superior results than previous methods, which contributes to improving student achievement and educational systems. The discussion of the various performance metrics confirms that the selected advanced models can be used to predict the dropout or non-dropout of students at the course level, even when the dataset is sparse.

In the advanced category, the AdaBoost classifier outperforms the XGB. The difference in accuracy between the prediction in the third quarter and the last quarter is approximately 3%. So, we recommend that educators pick the third quarter as the moment of intervention. It is also possible to explain the contribution of each feature in the prediction model with the AdaBoost classifier. The objective of which characteristics contribute the most to accurate predictions is personal; in this instance, they all proved positive, which is not always true.

Nevertheless, the presented study and its findings have some limitations. Due to the limited number of records, more sophisticated DL models, such as NN, could not be utilised. Frequently, this limitation can only be circumvented by employing the same precise research design employed in traditional educational technology research, in which a sufficient number of recordings is estimated or guaranteed prior to the experiment. The same remains true for the quantity and quality of individual features. In the learning process, repetition of certain activities is frequently employed. If these activities are dependent, their inclusion in the ML technique must be comprehensively scoped prior to data collection. However, more significant results could be anticipated if the course sections' activities were designed with the requirements of ML.

This study contributes to the solution of the problem of course-level student dropout. The results demonstrated that, despite a small sample size, adequately chosen indicators that do not require access to system logs can be advantageous when evaluating various performance metrics. In addition, the predictive models supplied information regarding the online learning environment activities and partial accomplishments of students.

We will continue collecting data from students enrolled in subsequent course cycles or other courses and implement the techniques mentioned above to generate a significant return. This endeavour enhances the performance of instructional frameworks, foundations, students, and instructors. In future studies, additional datasets will be applied to models of this type. In addition, considering the numerous known excellent classifiers, these results demonstrate the validity of the predictive models. Lastly, these models can assist teachers in comprehending students, identifying their deficiencies, developing learning styles, and combating dropout rates. It can also be advantageous for administrators to develop their instructional methods.

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